# Keep the Clock, Contrast the Prior: Training-Free Temporal Grounding for Video LLMs under a Visual-Token Budget

Anonymous CVPR submission

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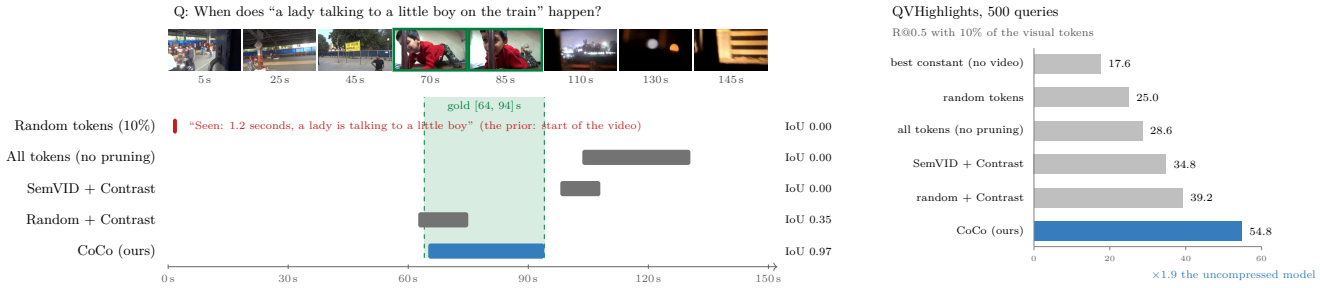



Figure 1. Temporal grounding from a pruned video memory. Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers "1.2 seconds"; QA-oriented pruning and the uncompressed model pick a later, wrong scene. CoCo keeps every frame anchored in time and contrasts the answer against the model's no-video prediction, and recovers the interval (IoU 0.97). Right: over 500 queries, CoCo nearly doubles the R@0.5 of the uncompressed model and is far above the best video-free constant.

## Abstract

Video LLMs are increasingly asked not only what happened but when: a grounded question such as "how long did I use the paper cutter?" needs an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding behaves differently. Four recent training-free pruners fall below random token selection on grounding, because a model reads time from the timestamps that survive in memory, and selection by saliency, uniqueness or text similarity removes whole stretches of the timeline. Where the memory is silent, the decoder fills the gap with a prior that does not depend on the video: asked with no video at all, the model still answers, starting intervals early and ending them too soon. We propose CoCo, a training-free method with two parts. Coverage gives every frame an equal token quota and fills it with the tokens the actual question attends to, so that every time anchor survives pruning. Contrast writes each time value against the model's own no-video prediction, with a strength that fades as the model grows confident, and reserves room for the final interval. Because grounding benchmarks carry strong positional biases, we evaluate every method against video-free floors. With 10% of the visual tokens, CoCo raises R@0.5 on QVHighlights from 25.0 (random tokens) and 28.6 (all tokens) to 54.8, where the best video-free constant reaches 17.6, and from 24.6 to 40.8 on S-EMBER grounded QA. We also report where it does not help: on Charades-STA and ActivityNet Captions the uncompressed model itself stays below the video-free floor, and no pruning method, ours included, rises above it.

## 1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts: a duration and the interval it comes from. Such grounded ques-

tions are central to egocentric assistants, video search and step-by-step guidance. They are hard for the same reason they are useful: the model must know when things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [32] spends about 330 visual tokens per pair of frames, so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget is therefore the default, and a fast-growing literature reports that 10–25% of the tokens preserve multiple-choice accuracy [15, 18, 23, 30, 33, 35]. This paper asks what pruning does to temporal grounding, and finds that the answer differs from QA in three ways.

Observation 1: QA pruners lose the timeline. Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. A frame whose visual tokens are all removed loses its timestamp, or keeps a timestamp with nothing to show. Pruners designed for QA remove frames that look redundant, unremarkable or unrelated to the words of the question, and those frames are exactly what makes a timeline usable for grounding. On S-EMBER grounded QA, all four recent pruners we evaluate fall below random token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.2). Random selection is a strong baseline for grounding because it keeps a few tokens on every frame.

Observation 2: where the memory is silent, a prior answers. Asked with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. A pruned memory shows the same habit: 30–56% of predicted intervals start in the first 5% of the window, against 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51%. The end of the interval suffers most: the median end error is two to three times the start error, and without video 87% of predicted ends fall more than half an interval too early (Sec. 3.3).

Observation 3: benchmark floors are high. Grounding benchmarks have strong positional regularities [31]. On S-EMBER, where questions are asked mid-stream and two thirds of the gold intervals end within the last fifth of the window, the best constant interval $[0.15q, q]$ reaches 36.9 mIoU, more than the uncompressed model (28.0). On ActivityNet Captions and Charades-STA the best constant also beats the uncompressed model. A claimed gain in grounding under compression must clear these floors, and we report them for every method (Sec. 3.4).

These observations suggest that a compressed memory for grounding needs two things that QA-oriented pruning does not provide: coverage, so that every part of the timeline keeps an anchor in memory, and a way to stop the decoder's prior from writing the interval where the evidence is thin. We propose CoCo (Coverage + Contrast), a training-free method with one component for each (Fig. 3):

- Coverage (Sec. 4.1). After the frames are encoded, the KV cache is pruned layer by layer. Every frame keeps an equal share of the layer's budget, and within each frame the tokens that the actual question attends to most are kept. Because a timestamp survives exactly when one of its frame's visual tokens survives, the equal quota keeps every time anchor, and the question decides what each anchor shows.
- Contrast (Sec. 4.2). While the answer is written, each time value is chosen by contrasting the model's prediction with its prediction from a video-free memory. The contrast acts only on time values, and its strength fades where the model is already confident. An answer reserve (Sec. 4.3) guarantees that the final interval is always written.

Neither operator is new in isolation: per-frame budgets appear in VidCom$^2$, KTV and AdaTP [2, 13, 30], question-attention KV eviction in SnapKV and PruneVid [23, 27], and contrast against a degraded input in VCD and M3ID [19, 26]. Our contributions are to show why temporal grounding needs this particular combination, and to make it work:

1. A diagnosis of grounding under compression (Sec. 3). QA-oriented pruners lose temporal coverage and fall below random selection; the decoder's video-free prior fills the gaps, mostly at the interval end; and benchmark floors are high enough to hide both effects.
2. CoCo (Sec. 4), which couples time-anchor coverage with prior-contrastive decoding of time values. It needs no training, no labelled calibration and no change to the model.
3. A floor-adjusted evaluation (Sec. 5) that reports, for every method, the best video-free constant interval, a length-rescaling baseline, and closed versus open intervals, with paired confidence intervals. All design choices were made on S-EMBER; the other three datasets were run once with the frozen configuration.
4. Results on four datasets (Sec. 6). With 10% of the tokens, CoCo exceeds the uncompressed model and every pruner on S-EMBER and QVHighlights, and clears the video-free floors by +7.6 mIoU on S-EMBER closed intervals and +22.1 on QVHigh-

lights, both significant. Question-driven selection improves as the budget shrinks, as long as each frame keeps enough tokens; at very small budgets, fewer frames with more tokens each recover what 64 frames lose. We also report the two datasets where the base model itself stays below the floor, and a broad set of negative results.

## 2. Related Work

Frame selection. BOLT [29] scores frames against the query with a contrastive image–text model and selects them by inverse transform sampling; AKS [34], MDP3 [6], FOCUS [3] and Q-Frame [7] optimise relevance together with coverage or diversity. All are evaluated on multiple-choice QA. Our optional retrieval front end (Sec. 4.4) is BOLT adapted to grounding, and we find that it contributes little once tokens are chosen by the question.

Visual token pruning for video LLMs. Query-agnostic methods compress by redundancy or saliency: FastVID [33] segments the video and keeps density peaks, VidCom$^2$ [30] gives each frame a budget by its uniqueness, FlashVID [18] selects by [CLS] attention and merges tokens across frames with trees, and DyCoke [35] and FrameFusion [21] merge similar tokens. Query-aware methods use the text: SemVID [15] scores tokens by word-embedding relevance with MMR diversity, AdaTP [2] boosts segments similar to the text, LGTTP [5] reads temporal words in the question to reallocate tokens, and PruneVid [23] keeps the top 40% of tokens by question attention at one LLM layer. KTV [13] is closest in structure (keyframes, then tokens with a relevance-weighted per-frame budget), and LGTTP is the only method we know that evaluates grounding, at a milder budget and offline. CoCo differs in three ways: the per-frame quota is equal by design, the score is the actual question's attention propagated exactly through all layers, and the quota exists to keep every timestamp, which we show matters for grounding.

KV-cache compression and video memory. SnapKV [27], $H_2O$ [39], PyramidKV [16] and Ada-KV [20] evict KV entries by attention in long-context LLMs; our token stage is SnapKV-style eviction applied to video with a per-frame constraint. For streaming video, HERMES [11] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks at question time, and StreamMem [9] and InfiniPot-V [4] maintain bounded memories. These works report QA only. Our memory is built per question from the frames up to the question time: it is causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

Contrastive decoding. VCD [26] contrasts the output distribution against a noised image, ICD [37] against a disturbed instruction, M3ID [19] against the absence of the image, and PAI [28] amplifies image attention while subtracting text-only logits; SEASON [8] extends the idea to video with temporally corrupted negatives. These methods target object hallucination or QA. We apply the no-input contrast to time values only, with a confidence-adaptive strength, and test whether its benefit for grounding is more than a uniform lengthening of the interval (Sec. 6.4).

Training-free temporal grounding. NumPro [38] draws frame numbers on the frames, FV-Action [10] scans the video with yes/no questions, REZE [14] recasts grounding as recognition, and [Authors to verify] [12] read intervals from calibrated attention heads, debiased against a blank-video run. All assume the full set of video tokens. CoCo addresses the complementary problem of grounding from a compressed memory; these readouts could run on top of it [TODO: run FV-Action and the attention readout on the same 10% memory].

Benchmark biases. Otani et al. [31] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because of positional priors in the annotations. We find the same on the datasets we use, and make video-free floors part of every comparison.

## 3. Grounding under a Token Budget: a Diagnosis

This section establishes the three observations of the introduction on S-EMBER, before any method is introduced. Unless stated otherwise, numbers in this section use random 10% pruning as the reference memory.

### 3.1. Setting

A question $x$ is asked at time $q$ about a video $v_{0:q}$. The model sees $F$=64 frames from $[0, q]$, encoded into a KV cache $\mathcal{M}$ of $N$ visual tokens interleaved with timestamp tokens (one "<t seconds>" per pair of frames in Qwen3-VL). A budget $\rho$ (default $\rho$=0.1) limits the number of tokens kept in each layer to $\rho N$, with kept timestamps counted inside the budget. The model answers with a short timeline:

Seen: $t_1$ seconds, …
Seen: $t_k$ seconds, …
Answer: …
Time: $[s, e]$

from which the interval $[s, e]$ is parsed. Grounding is scored by its IoU with the gold interval (mIoU and

Table 1. Where pruning keeps tokens. Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

| Selector | 0–10% | 10–25% | 25–50% | 50–75% | 75–100% | mIoU | Leak |
|---|---|---|---|---|---|---|---|
| HERMES | 4.7 | 3.9 | 2.6 | 2.5 | 27.6 | 18.3 | 49% |
| Stratified | 8.7 | 8.7 | 8.6 | 8.6 | 8.7 | 23.7 | 29% |
| Random | 8.9 | 8.8 | 8.6 | 8.7 | 8.6 | 25.8 | 30% |

R@{0.3, 0.5, 0.7}); grounded QA additionally requires the answer to be judged correct (GQ@0.5: correct answer and IoU $\geq 0.5$). We call an answer an early start when its interval starts in the first 5% of the window, and a leak when it starts early although the gold evidence starts later.

### 3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom$^2$, FlashVID and SemVID all score below random token selection (Sec. 6.2). The cause is visible in where the kept tokens lie (Tab. 1). HERMES's attention-plus-recency score keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere; FastVID keeps a token from only 25% of the sampled frames in a median question, because it retains one representative per run of similar frames, and runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. A timestamp is useful only when visual content of its frame survives with it, so a selector that empties frames leaves holes in the timeline. Holes hurt even when the score itself is reasonable: ranking HERMES's scores within each frame (an equal per-frame share, "stratified") instead of globally gains 5.3 mIoU [2.9, 7.6]*.

### 3.3. The prior fills the gaps

The model answers without video. Asked every question with an empty visual memory, the model still writes a timeline: 89% of its first Seen: moments are at 0 s and 90% of its intervals start in the first 5% of the window (88% leak). Its intervals are 0.45× the gold length (median) and 87% of its ends are more than half an interval too early. We call this the model's temporal prior: start early, stop soon.

The prior leaks into answers with video. The same signature appears in every memory, weaker (Fig. 2). About 30% of intervals start early at every random budget and with no pruning at all, against 16% of gold intervals, and the share rises when coverage is lost (HERMES 49%, HERMES streaming 37–39%). The leak is therefore not a matter of the number of tokens, but of where the memory is silent.

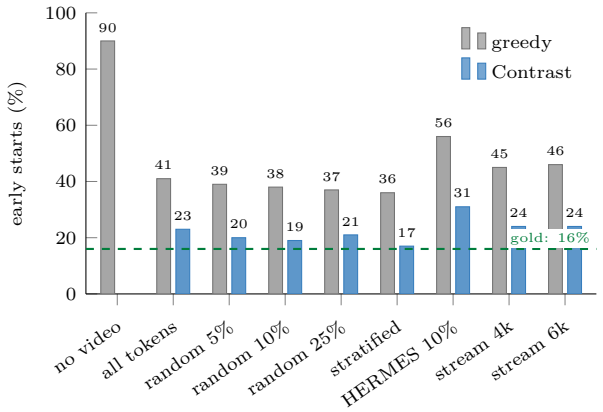



Figure 2. The temporal prior leaks into every memory, and Contrast removes most of it. Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). Contrast roughly halves it everywhere.

Removing the evidence causes the leak. To test causality we drop the uniform frames that fall inside the gold interval ("evidence") or the same number of frames elsewhere ("control"), keeping every other frame with its true timestamp (Tab. 2). With the evidence removed, the answers do not spread out: they move to the start of the video, and the leak rises by 14 points for both the unpruned and the random 10% memory. Within the HERMES 10% memory, questions whose evidence keeps under 5% of its tokens leak 56% of the time, against 33% when at least 15% survives.

An instruction does not remove it. Adding "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt leaves the leak almost unchanged (29.5% → 28.0%) and grounding unchanged (−0.1 mIoU). The prior acts in the output distribution and has to be removed there.

The end is the weak boundary. Across memories, the median end error is 0.40–0.69 of the gold length without contrastive decoding, against 0.14–0.30 for the start, and the most frequent failure is an interval entirely before the gold one (33% of all answers, Tab. 20).

### 3.4. Benchmark floors

Positional regularities make some intervals predictable without the video (Tab. 3). On S-EMBER, questions are asked while the stream is running and 67% of gold

Table 2. Causal test. Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. Contrast: adaptive PCD on time values (Sec. 4.2), no reserve.

| | | Greedy | | Contrast | |
|---|---|---|---|---|---|
| Memory | Removed | Acc./mIoU/GQ | Leak | Acc./mIoU/GQ | Leak |
| Random 10% | control | 16.6/28.8/8.8 | 31% | 15.4/34.6/10.5 | 11% |
| | evidence | 7.8/ 8.6/1.7 | 46% | 9.3/10.3/1.1 | 23% |
| All tokens | control | 19.6/29.4/10.1 | 37% | 18.1/35.9/12.6 | 15% |
| | evidence | 9.1/ 9.4/1.9 | 51% | 9.1/10.5/1.5 | 22% |

Table 3. Video-free floors. Best constant interval $[aq, bq]$ on a 20×20 grid, tuned on the test questions (each metric uses its own best constant; $[a, b]$ is the mIoU optimum), against the uncompressed model. Closed: share of questions with gold end $< 0.8q$.

| | Best constant | | | | | All tokens | |
|---|---|---|---|---|---|---|---|
| Dataset | $[a, b]$ | mIoU | R@0.5 | R@0.7 | Closed | mIoU | R@0.5 |
| S-EMBER | $[.15, 1]$ | 36.9 | 34.7 | 24.2 | 33% | 28.0 | 26.7 |
| QVHighlights | $[0, 1]$ | 26.8 | 17.6 | 8.8 | 62% | 29.1 | 28.6 |
| ActivityNet | $[0, .95]$ | 37.9 | 35.8 | 20.8 | 56% | 22.9 | 18.2 |
| Charades-STA | $[0, .45]$ | 29.7 | 29.6 | 15.4 | 70% | 16.6 | 14.2 |

intervals end within the last 20% of the window, so the constant $[0.15q, q]$ scores 36.9 mIoU. On ActivityNet Captions, whose moments cover 38% of the video on average, $[0, 0.95q]$ scores 37.9; on Charades-STA, $[0, 0.45q]$ scores 29.7. QVHighlights is the least biased (26.8 mIoU, 17.6 R@0.5). These constants are tuned on the test questions, so they upper-bound any video-free rule of this form. On three of the four datasets the best constant beats the uncompressed model, which makes the floors a necessary reference for every comparison that follows.

## 4. Method: CoCo

CoCo acts at the two places where Sec. 3 locates the failures: the memory (Coverage) and the decoder (Contrast). Neither needs training. Figure 3 gives an overview.

### 4.1. Coverage: equal quotas filled by the question

After the $F$ frames are encoded, the memory is pruned once, independently in each of the $L$ layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and $B$ the per-layer budget for visual tokens.

Question relevance. The question $x$ is appended to the full, uncompressed memory and propagated exactly through all layers, so that the queries at layer $\ell$ come from the real hidden states. The relevance of visual token $i$ at layer $\ell$ is the attention it receives from the question, averaged over heads $h$ and question positions $j$:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|} \sum_{h=1}^{H} \sum_{j \in x} \mathrm{softmax}_i\Big( \tfrac{1}{\sqrt{d}}\, Q_j^{(\ell,h)}\, K_i^{(\ell,h)^\top} \Big). \tag{1}$$

Equal quota. Every frame pair receives the same share of the budget, $B_f = \lfloor B/F' \rfloor$ (remainders assigned in order), and layer $\ell$ keeps

$$\mathcal{K}_\ell \;=\; \bigcup_{f=1}^{F'} \mathrm{top}_{B_f} \{\, a_\ell(i) \;:\; i \in \mathcal{T}_f \,\}. \tag{2}$$

The timestamp tokens of $f$ are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. With an equal quota every frame pair keeps at least one token at any practical budget, so every time anchor survives, and the question decides which part of each frame is shown. Kept tokens retain their original rotary positions, so their temporal order is preserved.

Why equal, and why the question. The equal quota is a deliberately simple coverage constraint. It guarantees the timeline, which grounding needs and multiple-choice QA does not, and it stops attention from collapsing onto a few frames: when quotas follow attention mass, attention sinks and recency bias give a single frame up to 227 of about 2,140 kept tokens. Relevance-weighted quotas with a floor, from SigLIP similarity or attention mass, did not beat equal quotas in a pre-registered test (−0.2 mIoU [−1.2, +0.8]). Using the actual question, rather than the generic probe questions of streaming memories such as HERMES [11], is what makes attention informative: generic probes with a global top-$k$ are the weakest variant we measured (18.3 mIoU).

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. Forcing one shared set (attention averaged over layers) costs CoCo −1.5 mIoU [−2.9, −0.0], whereas it does not change random selection (+0.7, n.s.). The benefit of per-layer selection is therefore specific to question-driven scores, and it does not explain why random selection beats input-level pruners.

Frames and budget. An equal quota needs enough tokens per frame to show anything: with 2–4 tokens per frame pair, the top-attended tokens carry little content and Coverage collapses (Sec. 6.6). When the budget is small, we therefore spend it on fewer frames: at a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. As a rule we use $F$=64 when each frame pair keeps at least ∼8 tokens and halve $F$ otherwise. [TODO: this rule was

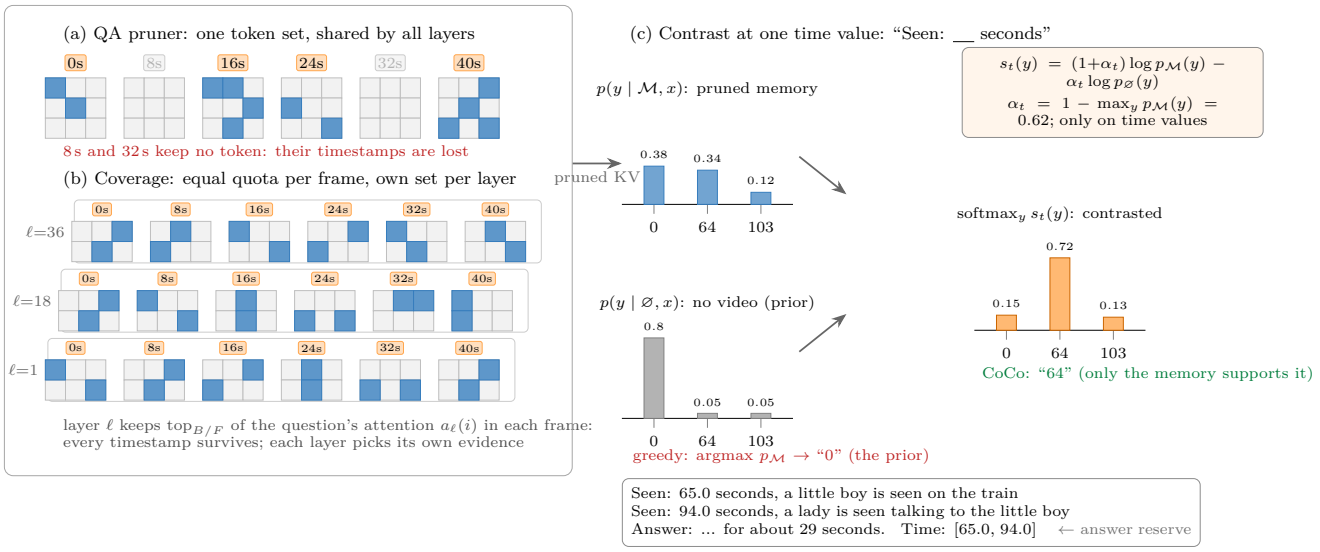



Figure 3. Overview of CoCo. (a) QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. (b) Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most in that layer. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 10). (c) Contrast at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. The answer reserve guarantees that the final interval is written (output shown is the real CoCo answer).

derived from the results of Sec. 6.6; validate it on held-out data.]

Cost. Coverage adds one forward pass of the question over the uncompressed memory per question. [TODO: measured latency and peak memory; the current implementation runs 3–4 scoring passes, reducible to one.]

### 4.2. Contrast: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and generated prefix but no visual tokens. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory carries no information about $y$, the evidence term is flat and the video-free prior decides: this is the failure of Sec. 3.3. Prior-contrastive decoding (PCD) amplifies the evidence term,

$$\begin{aligned} s_t(y) &= (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &= \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \qquad (4)$$

and picks $y_t = \arg\max_{y \in \mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with $\beta$=0.1, as in VCD [26], so that tokens the memory makes implausible are never promoted. Three choices adapt this rule to grounding.

No-video counterfactual. The negative memory is empty rather than noised or shuffled. It isolates exactly the prior measured in Sec. 3.3, and it is cheap: its cache holds only the prompt.

Time scope. The contrast is applied only while a time value is written: inside a Seen: time before its description and inside Time: $[s, e]$. All other tokens, including the answer text, are decoded greedily from $p_{\mathcal{M}}$, so the answer is unaffected where the memory already speaks.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\left(1 - \max_y p_{\mathcal{M}}(y)\right)$ with $\alpha_{\max}$=1. Where the memory makes the model confident (e.g. evidence that truly starts at 0 s), the contrast fades and the early answer is kept; where the model hesitates between times, the evidence term decides.

The timeline prompt matters: PCD acts on the listed Seen: moments and then on the interval, so the interval is written from contrasted observations. [TODO: report PCD with a direct-answer prompt, where it does not help.]

### 4.3. Answer reserve

Contrastive decoding lengthens the list of observed moments: under PCD the model sometimes keeps writ-

ing Seen: lines until the generation limit $T$, and never writes the interval. We reserve room for the answer: if $R$ tokens remain ($R$=64) and no Answer: line has started, the decoder writes "\nAnswer:" and continues normally. This is a format constraint, not a heuristic on the content: the interval is still chosen by PCD from the memory. It removes every unparseable output (from 25–68 of 500 to 0) and improves every PCD configuration on every dataset (Tab. 15).

### 4.4. Optional: retrieval-guided frames

When the window is much longer than $F$ frames, which frames are encoded can matter as much as which tokens are kept. As an optional front end we sample frames as in BOLT [29]. Each second of video is embedded once with SigLIP 2 [36], which gives a causal 1 fps index. The relevance $z_j$ of second $j \le q$ is its z-scored cosine similarity with the question, and the $F$ frames are taken at evenly spaced quantiles of

$$p(j) \;=\; \tfrac{1}{2}\cdot\tfrac{1}{q} \;+\; \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \qquad (5)$$

so that half of the frames stay uniform. Retrieval is skipped when uniform frames would be less than 1 s apart. Once tokens are chosen by the question, retrieval adds only +0.8 to +0.9 mIoU (not significant on S-EMBER, QVHighlights or ActivityNet), so we treat it as optional; the main-table runs include it.

## 5. Experimental Setup

Model. Qwen3-VL-8B-Instruct [32], 64 frames sampled uniformly from $[0, q]$ (or by Sec. 4.4) at native resolution, greedy decoding or PCD, at most 384 new tokens, batch size 8. All methods share the timeline prompt.

Datasets. S-EMBER grounded QA [1]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream; the memory contains only the frames before the question. QVHighlights [25] val: 500 queries with a single relevant window (a seed-0 sample of the 1,020 single-window queries), videos of about 150 s. ActivityNet Captions [24] val_1: 500 queries, one sentence per video, videos of about 115 s. Charades-STA [22] test: 500 queries, videos of about 30 s. For the last three, the question is "When does the following happen in the video: <query>?" and the window is the whole video. Our QVHighlights and ActivityNet copies are low-resolution re-encodes (153 and about 84 visual tokens per frame pair, against 660 on S-EMBER), so we also report budgets in absolute tokens.

Budget. 10% of the visual tokens of the 64-frame memory, timestamps included (S-EMBER: about 2,140 tokens; QVHighlights: about 520). ActivityNet is reported at 30% (400–900 tokens), because at 10% the memory holds only 120–300 tokens (Sec. 6.5).

Baselines. All tokens (no pruning); random tokens; FastVID [33], VidCom$^2$ [30], FlashVID [18] (vision-side stages) and SemVID [15], re-implemented and checked to keep the same tokens as the authors' code on test inputs; each with and without Contrast. [TODO: SnapKV-style observation-window eviction, PruneVid, global question top-$k$ without quotas, and FV-Action / attention readout on the same memory.]

Floor-adjusted evaluation. For every dataset we report three video-free or video-light references. (i) The best constant interval $[aq, bq]$ over a 20×20 grid, tuned on the test questions, which upper-bounds any constant rule. (ii) Length rescaling: the no-Contrast prediction stretched about its start by a factor fitted once on S-EMBER (1.76) and applied unchanged to the other datasets; it tests whether a gain could come from longer intervals alone. (iii) The split into closed questions (gold end $< 0.8q$) and the rest. End-snapping on S-EMBER is discussed in Appendix B.

Metrics and statistics. mIoU and R@$\{0.3, 0.5, 0.7\}$; on S-EMBER also answer accuracy and GQ@0.5, judged with the official S-EMBER judge prompt by a local Qwen3.8-27B-FP8 model, all rows of a table in one judging session. Answers that state times as minutes:seconds are converted to seconds for every method alike (Appendix A). We use a paired bootstrap over questions (2,000 resamples, 95% intervals) and mark with * the intervals that exclude 0. [TODO: video-clustered bootstrap.]

Protocol. All design choices (scores, quotas, $\alpha$, scope, reserve) were made on S-EMBER. The configuration was then frozen and run once on QVHighlights, ActivityNet and Charades-STA. [TODO: held-out S-EMBER videos as a test split.]

## 6. Results

We organise the results by question. Section 6.1 compares CoCo with all baselines and floors on four datasets. Section 6.2 shows how published pruners behave on grounding. Sections 6.3 and 6.4 analyse the two components. Sections 6.5 and 6.6 study the budget and how to spend it. Section 6.7 compares with upper bounds that use the gold interval, Sec. 6.8 analyses the optional retrieval stage, Sec. 6.9 lists what did not work, and Sec. 6.10 analyses the remaining errors.

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given. $\Delta$ is a paired difference with a 95% bootstrap interval; * marks intervals that exclude 0. Tables built from the early S-EMBER

phases (Tabs. 11 to 14, 18 and 20) predate the answer reserve and use the official parser and batch size 1; on S-EMBER the two parsers differ by at most 0.7 mIoU and batch size changes no paired result.

### 6.1. Main results

Tables 4 and 5 lead to three conclusions.

(i) On the least biased benchmark, CoCo nearly doubles the uncompressed model. On QVHighlights, with 10% of the tokens, CoCo reaches 54.8 R@0.5 (R@0.7: 30.2) against 28.6 (18.4) for all tokens and 25.0 (16.2) for random tokens, and clears every floor by a wide margin (+22.1 mIoU over the best constant, +8.6 over length rescaling).

(ii) On S-EMBER, the gain is real where the window end is not a shortcut. CoCo beats all tokens, random selection and SemVID under the same decoding. Against the floors its pooled gain is small and not significant, because two thirds of S-EMBER intervals end at the question time. On closed intervals, where that shortcut does not exist, CoCo gains +7.6 mIoU over the best constant and +7.5 over length rescaling.

(iii) On ActivityNet and Charades-STA, the base model is the bottleneck. The uncompressed model is far below the constant floor on both (22.9 vs. 37.9 and 16.6 vs. 29.7). CoCo improves on all tokens and on every pruner on ActivityNet, but no method reaches the floor, and on Charades-STA Coverage is worse than random selection (Sec. 6.6). We report these as limits, not successes.

Answers. CoCo also improves the answers (Tab. 6): accuracy 21.7 vs. 17.3 for random tokens (+4.4*) and 19.8 for all tokens, and GQ@0.5 13.3 vs. 7.4 and 8.8. Coverage alone gives the best accuracy (22.5); Contrast trades about one point of accuracy for its grounding gain, and differences below about two points of accuracy are within the decoding noise of the judge setup. [TODO: SemVID + Contrast with reserve in the same judging session.]

Question types. CoCo improves duration and counting questions most (Tab. 7). Location questions ("where did I put …") remain essentially unsolved by every method (mIoU $\leq$ 11): 62 of the 100 location questions have annotators who disagree on the interval (mean pairwise IoU $<$ 0.6), which caps any method.

### 6.2. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on S-EMBER (−2.3 to −7.6 mIoU, all significant), QVHighlights (−2.4 to −7.7; SemVID not significant) and Charades-STA (−2.8 to −5.2, all significant); on ActivityNet at 10% the input-level pruners are above random, which we discuss in Sec. 6.6.

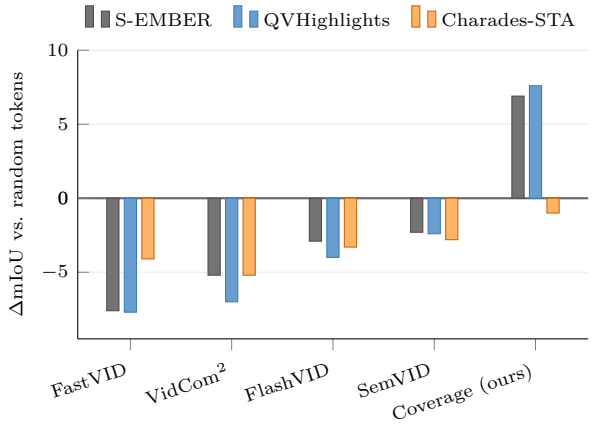



Figure 4. QA pruners fall below random selection on grounding. Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets (all significant except SemVID on QVHighlights). Coverage gains on the two long-video datasets and is level with random selection on 30 s Charades videos.

Random selection varies little across seeds (26.0 ± 0.6 mIoU over three seeds on S-EMBER), so the gaps are not noise. The ordering follows coverage (Tab. 1): the methods that empty frames most aggressively lose most, and those that spread tokens lose least. Accuracy, in contrast, is barely affected (Tab. 6): FastVID's accuracy is within noise of random selection (−2.1, n.s.) while its mIoU is 7.6 lower. A pruner built to keep answers keeps answers, not time. Contrast helps every pruner (+7 to +12 mIoU with the reserve) but does not close the gap to CoCo.

### 6.3. Analysis of Coverage

Contribution. Table 8 adds the components one at a time. Coverage is the largest gain on the memory side: +6.1* mIoU on S-EMBER and +6.8* on QVHighlights over random tokens. Retrieval adds +0.8 to +0.9 on top (not significant on any dataset); once tokens are chosen by the question, which frames are encoded matters less. [TODO: Coverage + Contrast without retrieval on all datasets.]

Design choices. Table 9 varies one choice at a time. The quota should be equal: five relevance-weighted variants, from SigLIP similarity, attention mass or both, with floors of 25–50%, are within ±0.5 mIoU of the equal quota and none is significant (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass are pulled by attention sinks and recency bias: a single frame received up to 227 of about 2,140 kept tokens. The score

Table 4. Grounding with 10% of the visual tokens (ActivityNet: 30%), mIoU / R@0.5. Grey rows are floors: the best video-free constant interval, tuned on the test questions, and the Retrieve + Coverage prediction without Contrast lengthened by a factor fitted on S-EMBER. Contrast includes the answer reserve. Coverage rows include optional retrieval, which the coverage gate disables on Charades-STA. Empty cells: not yet computed. Best non-floor result in bold.

| Method | S-EMBER | S-EMBER (closed) | QVHighlights | ActivityNet (30%) | Charades-STA |
|---|---|---|---|---|---|
| Best constant interval | 36.9 / 34.7 | 32.1 / 29.9 | 26.8 / 17.6 | 37.9 / 35.8 | 29.7 / 29.6 |
| Length rescaling of R + C | 38.3 / 40.0 | 32.3 / 33.1 | 40.2 / 42.4 | 34.0 / 33.0 | 24.0 / 17.6 |
| All tokens (no pruning) | 28.0 / 26.7 | 30.0 / 27.4 | 29.1 / 28.6 | 22.9 / 18.2 | 16.6 / 14.2 |
| Random tokens | 25.6 / 24.6 | 28.3 / 27.4 | 26.9 / 25.0 | 22.9 / 18.6 | 17.4 / 13.8 |
| FastVID [33] | 17.9 / 15.8 | | 19.2 / 15.4 | | 13.3 / 11.6 |
| VidCom$^2$ [30] | 20.4 / 18.7 | | 19.9 / 18.2 | | 12.3 / 8.4 |
| FlashVID [18] | 22.7 / 21.5 | | 22.8 / 20.4 | 20.5 / 15.4$^\dagger$ | 14.1 / 10.8 |
| SemVID [15] | 23.3 / 20.6 | | 24.5 / 21.2 | 22.8 / 18.6 | 14.6 / 11.2 |
| Random + Contrast | 35.0 / 33.9 | 36.8 / 38.2 | 37.8 / 39.2 | 31.4 / 29.4 | 26.2 / 24.0 |
| SemVID + Contrast | 34.8 / 35.8 | 34.3 / 34.4 | 34.8 / 34.8 | 30.2 / 27.8 | 21.7 / 19.2 |
| Coverage only | 32.5 / 34.1 | 33.0 / 35.7 | 34.5 / 35.6 | 25.4 / 21.2 | 16.4 / 13.6 |
| CoCo (Coverage + Contrast) | 39.3 / 40.8 | 39.7 / 43.3 | 48.8 / 54.8 | 34.7 / 32.4 | 23.3 / 21.2 |

$^\dagger$ at 10%.

Table 5. Paired gains of CoCo in mIoU (95% bootstrap interval; $^*$: excludes 0).

| CoCo vs. | S-EMBER | S-EMBER (closed) | QVHighlights | ActivityNet (30%) | Charades-STA |
|---|---|---|---|---|---|
| All tokens | +11.3 [TODO: CI] | | +19.7 [16.7, 22.8]$^*$ | +11.8 [9.3, 14.6]$^*$ | |
| Random + Contrast | +4.2 [1.7, 6.7]$^*$ | | +11.0 [8.3, 13.5]$^*$ | +3.3 [1.1, 5.5]$^*$ | −2.9 [−4.7, −1.1]$^*$ |
| SemVID + Contrast | +4.5 [1.9, 7.2]$^*$ | | +14.1 [11.2, 17.1]$^*$ | +4.5 [2.3, 6.7]$^*$ | |
| Best constant | +2.4 [−0.7, 5.6] | +7.6 [2.5, 12.6]$^*$ | +22.1 [18.7, 25.4]$^*$ | −3.1 [−5.9, −0.4]$^*$ | −6.4 [−9.5, −3.3]$^*$ |
| Length rescaling | +1.0 [−1.6, 3.7] | +7.5 [3.2, 11.7]$^*$ | +8.6 [5.5, 11.8]$^*$ | +0.8 [−2.0, 3.5] | −0.7 [−3.1, 1.6] |

Table 6. Grounded QA on S-EMBER (10%; one judging session). ΔAcc. vs. random tokens.

| Method | Acc. | GQ@0.5 | mIoU | ΔAcc. |
|---|---|---|---|---|
| All tokens | 19.8 | 8.8 | 28.0 | +2.5 |
| Random tokens | 17.3 | 7.4 | 25.6 | – |
| FastVID | 15.2 | 4.8 | 17.9 | −2.1 |
| VidCom$^2$ | 16.2 | 5.9 | 20.4 | −1.1 |
| FlashVID | 17.3 | 6.1 | 22.7 | +0.0 |
| SemVID | 19.8 | 7.6 | 23.3 | +2.5 |
| Coverage, uniform frames | 22.5 | 11.2 | 31.7 | +5.3$^*$ |
| Random + Contrast | 20.4 | 12.2 | 35.0 | +3.2 |
| SemVID + Contrast$^{\text{(no reserve)}}$ | 17.9 | 9.9 | 32.3 | +0.6 |
| CoCo | 21.7 | 13.3 | 39.3 | +4.4$^*$ |
| CoCo + coverage gate | 22.1 | 13.5 | 39.3 | +4.8$^*$ |

Table 7. S-EMBER by question type, mIoU / R@0.5 (10%).

| Method | Duration (191) | Counting (184) | Location (100) |
|---|---|---|---|
| All tokens | 37.6 / 40.3 | 30.1 / 25.5 | 5.9 / 3.0 |
| Random tokens | 33.2 / 35.1 | 29.6 / 26.1 | 3.5 / 2.0 |
| Coverage | 43.2 / 46.6 | 35.3 / 36.4 | 7.1 / 6.0 |
| Random + Contrast | 41.0 / 40.8 | 41.9 / 40.2 | 11.0 / 9.0 |
| SemVID + Contrast | 38.9 / 36.6 | 44.6 / 50.0 | 8.9 / 8.0 |
| CoCo | 47.6 / 50.8 | 46.9 / 48.4 | 9.3 / 8.0 |

Table 8. Component ladder (mIoU / R@0.5; ActivityNet at 30%). R: retrieval frames; C: Coverage; PCD: contrast; res.: answer reserve.

| Configuration | S-EMBER | QVHighlights | ActivityNet |
|---|---|---|---|
| Uniform, random tokens | 25.6 / 24.6 | 26.9 / 25.0 | 22.9 / 18.6 |
| R, random tokens | 29.4 / 29.5 | 28.7 / 28.0 | – |
| C (uniform frames) | 31.7 / 32.4 | 33.7 / 33.0 | 24.6 / 21.6 |
| R + C | 32.5 / 34.1 | 34.5 / 35.6 | 25.4 / 21.2 |
| R + C + PCD | 37.5 / 39.2 | 42.8 / 48.2 | – |
| R + C + PCD + res. (CoCo) | 39.3 / 40.8 | 48.8 / 54.8 | 34.7 / 32.4 |

should be the actual question: HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without. Deterministic top-$k$ beats sampling from a coverage-floored density (31.7 vs. 30.0); with the quota in place, coverage is already guaranteed. A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

Per-layer selection. Table 10 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, which rules out per-

Table 9. Design choices of Coverage (S-EMBER, greedy decoding). Frames: R = retrieval, U = uniform. Δ vs. the equal quota with question top-$k$ on the same frames.

| Token selection | Frames | mIoU | Δ |
|---|---|---|---|
| Equal quota, question top-$k$ | R | 32.5 | – |
| Quota ∝ SigLIP relevance (floor 50%) | R | 33.0 | +0.5 [−0.5, 1.4] |
| Quota ∝ SigLIP (floor 25%) | R | 32.6 | +0.1 [−1.2, 1.3] |
| Quota ∝ SigLIP + attention | R | 32.9 | +0.4 [−0.7, 1.5] |
| Quota ∝ attention mass | R | 32.1 | −0.4 |
| Contrastive token score‡ | R | 33.0 | +0.5, n.s. |
| One token set shared by all layers | R | 31.9 | −0.6 [−2.3, 1.1] |
| Equal quota, question top-$k$ | U | 31.7 | – |
| Quota ∝ SigLIP relevance | U | 31.4 | −0.2 [−1.2, 0.8] |
| Equal quota, sampling instead of top-$k$ | U | 30.0 | −1.7 |
| Contrastive token score‡ | U | 30.5 | −1.2 |
| HERMES score, equal quota (stratified) | U | 23.7 | |
| HERMES score, global top-$k$ | U | 18.3 | |
| Question, global top-$k$, no quota | U | [TODO: ] | |

‡ $\log a(\text{question}) - \log a(\text{generic sentence})$.

Table 10. Per-layer vs. shared token sets (S-EMBER, 10%). Δ = shared − per layer.

| Method | Per layer | Shared | ΔmIoU |
|---|---|---|---|
| Random tokens | 25.6 | 26.3 | +0.7 [−0.7, 2.2] |
| Random + Contrast | 35.0 | 35.8 | +0.7 [−0.9, 2.3] |
| R + C | 32.5 | 31.9 | −0.6 [−2.3, 1.1] |
| CoCo | 39.3 | 37.8 | −1.5 [−2.9, −0.0]* |

layer diversity as the reason random selection beats input-level pruners. It costs CoCo −1.5* mIoU: different layers attend to different tokens, and question-driven selection benefits from letting each choose.

### 6.4. Analysis of Contrast

Contrast is a plug-in for any memory. Table 11 applies Contrast to eight memories: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. It improves mIoU significantly on all eight (+4.6 to +8.0) and GQ@0.5 on six, roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection it already beats the uncompressed model (32.6 vs. 28.0). Contrast also transfers to FastVID (+7.6* mIoU, leak 38% → 12%), the pruner that loses coverage most.

The no-video counterfactual is the right one. Table 12 changes only what the memory is contrasted against. The no-video input isolates the prior and removes it (early starts 38% → 19%). Noised frames, as in VCD, still show part of the video and still produce the prior, so the two runs disagree mainly about content and little of the prior is removed (+0.5 mIoU). Permuted timestamps leave the prior in place as well. A one-sided rule that only penalises tokens the video argues against is worse than PCD (−5.6* and −4.8*): without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, leaves the prior in place.

Time scope and adaptive strength. Contrasting every token also changes words that have nothing to do with time: the answer text changed in 85% of questions, lists of more than eight moments rose from 12 to 56, and over-counting rose from 14% to 34% of counting questions. Restricting the contrast to time values keeps the grounding gain and disturbs the text less (Tab. 13, top). Contrast also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7; with the adaptive strength, which fades when the model is confident, it rises to 23.1 (Tab. 13, bottom; +6.4* vs. constant). The adaptive rule also gives the best discrimination, early starts on early-evidence questions minus early starts on the others (56.4 vs. 49.6 for greedy, all-tokens memory).

PCD is near the limit of decoding. We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 14). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones. The remaining early-start errors are cases where the memory itself is unsure; they need evidence, not a better decoding rule.

Is Contrast more than a longer interval? Contrast lengthens predicted intervals (median predicted/gold length 0.57 → 0.67 on S-EMBER), and longer intervals alone raise mIoU on benchmarks with long gold intervals. The length-rescaling floor tests this directly (Tab. 5, last row). On QVHighlights (+8.6*) and on S-EMBER closed intervals (+7.5*), CoCo clearly beats rescaling: Contrast moves the interval toward the evidence, not just outward. On pooled S-EMBER, ActivityNet and Charades-STA the two are statistically tied, so on those sets the benefit of Contrast cannot be separated from a calibrated lengthening. [TODO: start/end error decomposition on closed questions; mismatched-video and shuffled-frame negatives.]

Answer reserve. Contrast rewards fine time steps that the prior finds unlikely, so it sometimes lists one continuing activity at one-second steps until the token limit. The reserve removes every such failure and improves every PCD configuration we tested (Tab. 15), most where run-ons were frequent (SemVID + PCD on Charades, CoCo on QVHighlights). It costs nothing when the model answers within the limit, which every greedy

Table 11. Contrast on eight memories (S-EMBER, Acc. / mIoU / GQ@0.5; no answer reserve). PCD: plain contrast on all tokens ($\alpha$=0.5); final: confidence-adaptive strength on time values only. $\Delta$ final vs. greedy.

| Memory | Greedy | PCD | Final | $\Delta$mIoU | $\Delta$GQ@0.5 | Early starts |
|---|---|---|---|---|---|---|
| Random 5% | 15.8 / 23.6 / 5.9 | 15.4 / 29.2 / 9.1 | 16.4 / 31.5 / 9.9 | +8.0* | +4.0* | 39% → 20% |
| Random 10% | 15.2 / 25.8 / 7.2 | 16.2 / 32.1 / 10.1 | 16.2 / 32.6 / 10.1 | +6.8* | +2.9 | 38% → 19% |
| Random 25% | 17.7 / 28.2 / 8.2 | 18.9 / 33.0 / 11.8 | 18.9 / 34.9 / 13.3 | +6.7* | +5.1* | 37% → 21% |
| All tokens | 18.5 / 28.0 / 8.6 | 19.8 / 31.5 / 11.2 | 21.5 / 32.7 / 13.3 | +4.6* | +4.6* | 41% → 23% |
| HERMES 10% | 13.1 / 18.3 / 4.0 | 15.4 / 22.6 / 7.2 | 13.3 / 24.1 / 5.9 | +5.7* | +1.9 | 56% → 31% |
| Stratified 10% | 14.1 / 23.7 / 6.1 | 14.5 / 29.9 / 8.8 | 15.8 / 31.0 / 10.5 | +7.3* | +4.4* | 36% → 17% |
| HERMES streaming 4k | 13.5 / 21.6 / 4.0 | 17.1 / 27.4 / 8.2 | 15.4 / 28.6 / 8.2 | +6.9* | +4.2* | 45% → 24% |
| HERMES streaming 6k | 14.3 / 22.6 / 4.8 | 14.7 / 28.1 / 7.6 | 15.4 / 29.8 / 8.4 | +7.2* | +3.6* | 46% → 24% |
| Mean of 8 | 15.3 / 24.0 / 6.1 | 16.5 / 29.2 / 9.3 | 16.6 / 30.7 / 10.0 | +6.7 | +3.9 | |

Table 12. What to contrast against (S-EMBER, random 10%). Only the counterfactual input changes; the main input is always the real memory.

| Counterfactual | mIoU | GQ@0.5 | Early |
|---|---|---|---|
| None (greedy) | 25.8 | 7.2 | 38% |
| No video (ours) | 32.1 (+6.2*) | 10.1 (+2.9*) | 19% |
| Noised frames (VCD [26]) | 26.3 (+0.5) | 7.8 | 30% |
| Permuted timestamps | +0.9 (n.s.) | – | – |
| Evidence-against rule, $\alpha$=1 / 2 | 26.5 / 27.3 | – | – |
| Attention gain (PAI-style) | +1.3 (n.s.) | +0.8 | 42% |
| Instruction "do not assume early" | 25.7 | 6.9 | 37% |

Table 13. Time scope and adaptive strength (S-EMBER, Acc. / mIoU / GQ@0.5). Top: contrast on all tokens vs. time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window.

| Memory | All tokens | Time values only |
|---|---|---|
| Random 10% | 16.2 / 32.1 / 10.1 | 16.0 / 32.1 / 10.3 |
| Random 25% | 18.9 / 33.0 / 11.8 | 17.7 / 34.0 / 11.8 |
| All tokens | 19.8 / 31.5 / 11.2 | 20.8 / 32.9 / 12.4 |
| HERMES 10% | 15.4 / 22.6 / 7.2 | 14.7 / 24.3 / 7.4 |
| Streaming 4k | 17.1 / 27.4 / 8.2 | 16.2 / 29.2 / 9.3 |
| Early-evidence questions (78), all tokens memory | | |
| Greedy | 21.8 / 49.2 / 16.7 | |
| PCD, constant $\alpha$ | 16.7 / 55.3 / 12.8 | |
| PCD, adaptive $\alpha$ | 23.1 / 56.9 / 19.2 | |

run does.

Prompt format. Contrast needs a reasoning-style answer: with the official direct-answer prompt it does not help (+0.4 mIoU). An event timeline that lists start and end per occurrence (Event: 12.0 – 15.5 seconds) improves the greedy baseline by +7.4* mIoU by stating extents, and Contrast still adds +2.1 to +3.3 on top (35.0 mIoU, 12.6 GQ@0.5 at 768 tokens). The two fixes overlap: the format corrects extents, Contrast corrects the leak. CoCo uses the simpler timeline in all main results. [TODO: event timeline with CoCo.]

Table 14. Decision rules replayed at the first time value (S-EMBER, random 10%). Share of answers starting at "0".

| Rule | Early evidence ↑ | Other (leak) ↓ |
|---|---|---|
| Greedy | 69% | 16% |
| PCD, $\alpha$=0.5 | 46% | 6% |
| PCD, adaptive | 51% | 7% |
| Evidence-against, $\alpha$=4 | 51% | 9% |
| Skewed or tempered prior | 50–55% | 8–11% |
| Best threshold (uses gold) | 46% | 6% |

Table 15. Answer reserve (paired $\Delta$mIoU) and unparseable answers before → after.

| Configuration | S-EMBER | QVHighlights | Charades |
|---|---|---|---|
| Random + PCD | +1.9* | +1.5* | +2.7* |
| SemVID + PCD | +2.5* | – | +5.6* |
| R + C + PCD | +1.8* | +6.1* | +2.4* |
| Unparseable (of 500) | 26–39 → 0 | 25–60 → 0 | 28–68 → 0 |

## 6.5. Budget

Table 16 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection is flat or even improves as the budget shrinks: on S-EMBER, Retrieve + Coverage rises from 31.0 (20%) to 34.6 (5%) and CoCo from 36.8 to 39.3, while random selection and SemVID degrade. Pruning by the question removes distractors as well as cost: at 5% CoCo leads SemVID + Contrast by +7.4* mIoU, at 20% by +1.5 (n.s.). Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair, the top-attended tokens carry little content, and CoCo collapses (9.7 and 11.6), below random selection and

Table 16. Budget sweep at 64 frames (mIoU). The second row gives the number of kept tokens per memory.

| | S-EMBER | | | QVHighlights | | | ActivityNet | | |
|---|---|---|---|---|---|---|---|---|---|
| Budget | 5% | 10% | 20% | 5% | 10% | 20% | 10% | 30% | 50% |
| Kept tokens | ∼1,070 | ∼2,140 | ∼4,280 | ∼260 | ∼520 | ∼1,040 | 120–300 | 400–900 | 600–1,500 |
| Random | 23.9 | 25.6 | 27.4 | | 26.9 | | 17.8 | 22.9 | 23.5 |
| SemVID | 20.9 | 23.3 | 26.5 | | 24.5 | | 21.2 | 22.8 | 22.3 |
| R + C (no Contrast) | 34.6 | 32.5 | 31.0 | | 34.5 | | 11.3 | 25.4 | 25.0 |
| SemVID + Contrast | 31.9 | 34.8 | 35.3 | | 34.8 | | 26.8 | 30.2 | 31.4 |
| Random + Contrast | 34.5 | 35.0 | 36.5 | | 37.8 | | 21.5 | 31.4 | 32.2 |
| CoCo | 39.3 | 39.3 | 36.8 | 9.7 | 48.8 | 49.4 | 11.6 | 34.7 | 34.4 |

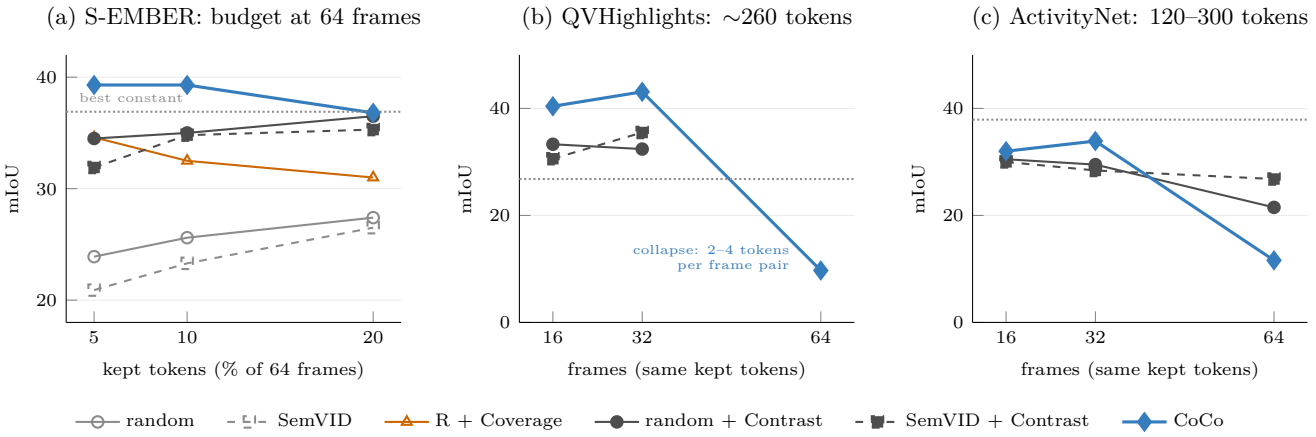



Figure 5. Budget and how to spend it. (a) At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; CoCo leads at every budget. (b, c) At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and CoCo collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9). Dotted grey: best video-free constant.

below SemVID, whose cross-frame merging degrades more gracefully. The threshold is absolute, not relative: it depends on the number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies.

### 6.6. Fewer frames or fewer tokens per frame?

The collapse at small budgets is not a limit of the equal quota itself but of how the budget is spent. In Tab. 17 every row of a block keeps the same number of tokens and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and CoCo recovers from 9.7 to 43.1 mIoU on QVHighlights and from 11.6 to 33.9 on ActivityNet, where it now beats SemVID + Contrast (+5.5) and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little: past a point, temporal coverage becomes the bottleneck. The baselines gain from fewer frames too, but less. On Charades-STA (30 s videos) fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1: keep at least about eight tokens per frame pair and reduce $F$ when the budget does not allow it. At equal tokens, more frames with fewer tokens each does not help either: 128 frames at 5% on S-EMBER give +1.2 mIoU (n.s.) without Contrast and −3.6* with it, with 74–78 unparseable answers.

### 6.7. Comparison with gold-informed upper bounds

Table 18 gives the memory information it should not have. At the same budget, Contrast without any gold information beats perfect token selection on grounding (+4.7* mIoU over the oracle selector at 10%, +4.5* at 5%): knowing which tokens to keep does not stop the prior, and the oracle selector still leaks 33% of the time. Against the strongest oracle, which sees only the evidence, Contrast closes most of the grounding gap (GQ@0.5 13.3 vs. 13.9, n.s.), and the remaining gap is in the answers (accuracy 21.5 vs. 26.3*). Even the oracle window leaks 45%: with every frame inside the evidence, the model still often starts near 0 s. More

Table 17. Spending a fixed token budget on fewer frames (mIoU / R@0.5, all with Contrast and reserve). Within each block every row keeps the same number of tokens: $F$ frames at keep ratio $\rho \cdot 64/F$.

| Dataset (kept tokens) | Frames × keep ratio | Random + Contrast | SemVID + Contrast | CoCo |
|---|---|---|---|---|
| QVHighlights (∼260) | 64 × 5% | – | – | 9.7 / 6.6 |
| | 32 × 10% | 32.4 / 29.2 | 35.5 / 33.6 | 43.1 / 46.6 |
| | 16 × 20% | 33.3 / 32.4 | 30.6 / 28.0 | 40.4 / 41.0 |
| ActivityNet (120–300) | 64 × 10% | 21.5 / 18.2 | 26.8 / 23.6 | 11.6 / 7.6 |
| | 32 × 20% | 29.5 / 26.8 | 28.4 / 25.8 | 33.9 / 32.0 |
| | 16 × 40% | 30.5 / 26.8 | 30.0 / 27.6 | 32.0 / 29.8 |
| Charades-STA (∼10% of 64) | 64 × 10% | 26.2 / 24.0 | 21.7 / 19.2 | 23.3 / 21.2 |
| | 32 × 20% | 24.6 / 22.0 | 20.8 / 16.0 | 23.6 / 21.0[§] |

[§] Coverage on uniform frames; retrieval is gated off on Charades-STA.

Table 18. Upper bounds that use the gold interval (S-EMBER, greedy unless noted; Contrast without reserve). Oracle selector: the gold frames' tokens are kept first. Oracle window: all 64 frames inside the gold interval.

| Memory | Acc. | mIoU | R@0.5 | GQ | Leak |
|---|---|---|---|---|---|
| No video | 10.5 | 7.9 | 4.8 | 0.8 | 88% |
| Random 10% | 15.2 | 25.8 | 24.2 | 7.2 | 30% |
| Random 10% + Contrast | 16.2 | 32.6 | 33.1 | 10.1 | 11% |
| Oracle selector 10% | 17.5 | 27.9 | 25.7 | 8.4 | 33% |
| Gold-window zoom 10% | 14.5 | 28.2 | 26.1 | 7.8 | 25% |
| All tokens | 18.5 | 28.0 | 26.7 | 8.6 | 33% |
| All tokens + Contrast | 21.5 | 32.7 | 34.1 | 13.3 | 13% |
| Oracle window | 26.3 | 35.3 | 36.2 | 13.9 | 45% |

evidence alone does not remove the prior; the two levers are complementary.

### 6.8. Analysis of the optional retrieval

The SigLIP 2 index does find the evidence: over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663, and the top-ranked second lies in the gold interval for 63% of questions (chance 36%) and 41% of location questions (chance 13.5%). Retrieval-guided frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens (+3.8* mIoU) but adds only +0.8 to +0.9 (n.s.) once tokens are chosen by the question, because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666), as expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [36]. Re-weighting the token budget of uniform frames by the same relevance adds +1.3 (n.s.): which frames are encoded matters more than how the budget is split among fixed frames.

Table 19. Negative results (S-EMBER; ΔmIoU against the stated reference).

| Idea | Reference | ΔmIoU |
|---|---|---|
| VCD-style noised-frame contrast | greedy | +0.5 |
| Permuted-timestamp contrast | greedy | +0.9 |
| Evidence-against decoding rule | PCD | −5.6* |
| Attention gain while decoding | greedy | +1.3 |
| Bias stated in the instruction | greedy | −0.1 |
| Relevance-guided frame zoom | greedy | −1.5 |
| Zoom inside the gold window | greedy | +2.3 |
| Unequal per-frame quotas (5 variants) | equal quota | −0.4 to +0.5 |
| Contrastive token score | question top-$k$ | −1.2 to +0.5 |
| Prior-subtracted retrieval | SigLIP | AUROC ≤ +0.003 |
| 128 frames at the same budget | 64 frames | +1.2 / −3.6* |

Table 20. Error types on S-EMBER, pooled over eight memories (3,800 answers), greedy vs. Contrast (no reserve).

| Outcome | Greedy | Contrast | Δ |
|---|---|---|---|
| IoU ≥ 0.5, answer correct | 6.1% | 9.9% | +3.8* |
| IoU ≥ 0.5, answer wrong | 16.3% | 20.7% | +4.4* |
| Partial overlap (IoU < 0.5) | 17.0% | 15.7% | −1.3 |
| Inside gold, too short | 22.2% | 18.5% | −3.7* |
| Entirely before gold | 33.1% | 18.9% | **−14.2*** |
| Entirely after gold | 2.7% | 5.7% | +3.0* |
| Covers gold, too long | 2.2% | 2.5% | +0.3 |
| No answer / unparsed[¶] | 0.5% | 8.1% | +7.6* |

[¶] removed by the answer reserve (Tab. 15).

### 6.9. What did not work

Table 19 collects the ideas that gave little or nothing. They share a lesson: at a fixed budget, redistributing evidence (zooming, re-weighting frames, unequal quotas) is close to its ceiling, and only removing the prior where it acts (Contrast) or guaranteeing coverage (the equal quota) gives large gains.

### 6.10. Where answers still fail

Table 20 shows that Contrast works where it was aimed: intervals entirely before the gold one, the signature of the prior, fall from the largest error type to less than a fifth of answers. The remaining failures point to four limits. (i) Answers, not intervals: twice as many answers are well grounded but wrong as grounded and right; counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) Recent events: 81% of the intervals that still end before the gold one have evidence in the second half of the window, where an earlier similar event is picked; GQ@0.5 is 4.4 when the evidence lies in the last third of the window. (iii) Sparse evidence: with fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more; no decoding rule recovers evidence that is not in the input. (iv) Label ambiguity: on the 98 questions whose annotators disagree (mean pairwise IoU $< 0.6$), every method scores near zero.

## 7. Limitations

Base-model floors. On ActivityNet Captions and Charades-STA, Qwen3-VL-8B with 64 frames stays below the best constant interval even without pruning, and so does every compressed variant. Our claims on these datasets are relative to the model.

Minimum tokens per frame. Equal quotas fail when a frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.6), but the rule we use (at least about eight tokens per frame pair) was derived from the same results and still needs held-out validation.

Short, densely sampled videos. When uniform frames are less than a second apart, spreading tokens at random beats question-driven selection.

Not a streaming memory. The memory is rebuilt per question from the frames before the question. It is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token are those of the uncompressed model: CoCo reduces the memory the decoder reads, not the cost of building it. Contrast also adds a second, prompt-only decoding pass for time-value tokens. [TODO: cost table; write-once memory re-ranked at question time.]

Scope of the evaluation. All results use one model, 475–500-question subsets, and a local judge with the official prompt. Every hyperparameter was chosen on the S-EMBER questions; the other datasets were run once with the frozen configuration, but a held-out S-EMBER split is still to be evaluated.

## 8. Conclusion

Temporal grounding asks a compressed video memory for something multiple-choice QA does not: a complete timeline. Pruners designed for QA remove parts of it and fall below random selection, and the decoder's video-free prior fills the gaps, mostly by ending intervals too early. CoCo keeps every time anchor with an equal, question-filled quota and writes time values against the model's own no-video prediction. With a tenth of the visual tokens it grounds better than the uncompressed model on S-EMBER and QVHighlights, and it clears video-free floors wherever the base model does. The floors are a lesson of their own: on two standard benchmarks a constant interval beats the uncompressed model, and grounding results under compression should be read against them.

## References

[1] [authors to verify]. S-EMBER: grounded streaming video question answering. [cite: S-EMBER benchmark paper]. 7

[2] [authors to verify]. AdaTP: Attention-debiased token pruning for video large language models. arXiv preprint arXiv:2505.20100, 2025. 2, 3

[3] [authors to verify]. FOCUS: Efficient keyframe selection for long video understanding. arXiv preprint arXiv:2510.27280, 2025. 3

[4] [authors to verify]. InfiniPot-V: Memory-constrained KV cache compression for streaming video understanding. arXiv preprint arXiv:2506.15745, 2025. 3

[5] [authors to verify]. Language-guided temporal token pruning for efficient VideoLLM processing. In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025. [verify]. 3

[6] [authors to verify]. MDP3: A training-free approach for list-wise frame selection in video-LLMs. arXiv preprint arXiv:2501.02885, 2025. 3

[7] [authors to verify]. Q-Frame: Query-aware frame selection and multi-resolution adaptation for video-LLMs. In IEEE/CVF International Conference on Computer Vision (ICCV), 2025. [verify]. 3

[8] [authors to verify]. SEASON. arXiv preprint arXiv:2512.04643, 2025. [verify title]. 3

[9] [authors to verify]. StreamMem: Query-agnostic KV cache memory for streaming video understanding. arXiv preprint arXiv:2508.15717, 2025. 3

[10] [authors to verify]. Your VLM already knows when. arXiv preprint arXiv:2608.08315, 2026. [verify title]. 3

[11] [authors to verify]. HERMES. In Annual Meeting of the Association for Computational Linguistics (ACL), 2026. [verify title]. 3, 5

[12] [Authors to verify]. MLLMs know when before speaking. arXiv preprint arXiv:2605.21954, 2026. [verify title]. 3

[13] [authors to verify]. Keyframes and key tokens selection for efficient training-free video LLMs. arXiv preprint arXiv:2602.03615, 2026. 2, 3

[14] [authors to verify]. REZE. arXiv preprint arXiv:2608.04480, 2026. [verify title]. 3

[15] [authors to verify]. SemVID. In European Conference on Computer Vision (ECCV), 2026. arXiv:2603.05663; [verify title]. 2, 3, 7, 9

[16] Zefan Cai et al. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling. arXiv preprint arXiv:2406.02069, 2024. 3

[17] Shangzhe Di et al. Streaming video question-answering with in-context video KV-cache retrieval. In International Conference on Learning Representations (ICLR), 2025. 3

[18] Ziyang Fan et al. FlashVID: Training-free tree-based spatiotemporal token merging for video large language models. In International Conference on Learning Representations (ICLR), 2026. arXiv:2602.08024; [verify]. 2, 3, 7, 9

[19] Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. Multi-modal hallucination control by visual information grounding. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 2, 3

[20] Yuan Feng et al. Ada-KV: Optimizing KV cache eviction by adaptive budget allocation for efficient LLM inference. arXiv preprint arXiv:2407.11550, 2024. 3

[21] Tianyu Fu et al. FrameFusion: Combining similarity and importance for video token reduction on large visual language models. arXiv preprint arXiv:2501.01986, 2025. [verify]. 3

[22] Jiyang Gao, Chen Sun, Zhenheng Yang, and Ram Nevatia. TALL: Temporal activity localization via language query. In IEEE International Conference on Computer Vision (ICCV), 2017. 7

[23] Xiaohu Huang, Hao Zhou, and Kai Han. PruneVid: Visual token pruning for efficient video large language models. In Findings of the Association for Computational Linguistics (ACL), 2025. 2, 3

[24] Ranjay Krishna, Kenji Hata, Frederic Ren, Li Fei-Fei, and Juan Carlos Niebles. Dense-captioning events in videos. In IEEE International Conference on Computer Vision (ICCV), 2017. 7

[25] Jie Lei, Tamara L. Berg, and Mohit Bansal. QVHighlights: Detecting moments and highlights in videos via natural language queries. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 7

[26] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 2, 3, 6, 11

[27] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In Advances in Neural Information Processing Systems (NeurIPS), 2024. 2, 3

[28] Shi Liu, Kecheng Zheng, and Wei Chen. Paying more attention to image: A training-free method for alleviating hallucination in LVLMs. In European Conference on Computer Vision (ECCV), 2024. 3

[29] Shuming Liu et al. BOLT: Boost large vision-language model without training for long-form video understanding. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. arXiv:2503.21483. 3, 7

[30] Xuyang Liu et al. Video compression commander: Plug-and-play inference acceleration for video large language models. In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025. arXiv:2505.14454. 2, 3, 7, 9

[31] Mayu Otani, Yuta Nakashima, Esa Rahtu, and Janne Heikkilä. Uncovering hidden challenges in query-based video moment retrieval. In British Machine Vision Conference (BMVC), 2020. 2, 3

[32] Qwen Team. Qwen3-VL technical report, 2025. [verify]. 2, 7

[33] Leqi Shen et al. FastVID: Dynamic density pruning for fast video large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2025. arXiv:2503.11187. 2, 3, 7, 9

[34] Xi Tang et al. Adaptive keyframe sampling for long video understanding. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. arXiv:2502.21271. 3

[35] Keda Tao et al. DyCoke: Dynamic compression of tokens for fast video large language models. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [verify]. 2, 3

[36] Michael Tschannen et al. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. arXiv preprint arXiv:2502.14786, 2025. 7, 13

[37] Xintong Wang, Jingheng Pan, Liang Ding, and Chris Biemann. Mitigating hallucinations in large vision-language models with instruction contrastive decoding. In Findings of the Association for Computational Linguistics (ACL), 2024. 3

[38] Yongliang Wu et al. Number it: Temporal grounding videos like flipping manga. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [verify]. 3

[39] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2023. 3

## A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, e.g. Time: [1:45, 2:07], which the official S-EMBER parser scores as missing. We convert h:mm:ss and m:ss to seconds for every method alike. Method rankings are unchanged with the official parser. The main numbers with the official parser are: QVHighlights, CoCo 44.6 / 50.2 (vs. 48.8 / 54.8); ActivityNet at 30%, 33.3 / 31.0 (vs. 34.7 / 32.4). [TODO: full table with the official parser.]

## B. End-snapping on S-EMBER

Setting every predicted end to the question time $q$ raises every S-EMBER method to 42–46 mIoU, more than the 39.3 of CoCo, because 67% of the gold intervals end within the last fifth of the window. On closed intervals, end-snapping hurts every method (to 22–24 mIoU) while CoCo keeps 39.7; on QVHighlights it lowers CoCo from 48.8 to 40.5. A decoder that decides whether an interval is still open at question time is a natural extension. [TODO: open-interval decoding.]

## C. Reproduction

CoCo corresponds to the following overrides of the HERMES code base:

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time run.answer_reserve=64
(optional retrieval) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```

Floors and paired intervals are computed by docs/paper/analysis/floors.py.